Prodigy Anesthesia Quick Review

Asthma

List the clinical manifestations of asthma.	Dyspnea, eosinophilia, nonproductive or productive cough, wheezing, chest tightness *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 182.*
Define status asthmaticus.	Status asthmaticus is a life-endangering bronchospasm that is unresponsive to treatment. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 186.*
What are the two components to the treatment of asthma?	1. Treatments that aim to "control" the airway so that acute airway narrowing is less frequent. This treatment includes theophylline, systemic and inhaled corticosteroids, and antileukotrienes. 2. Treatments that aim to "relieve" or rescue acute bronchospasm, such as beta-adrenergic agonists and anticholinergic drugs. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 185.*
What are the three main characteristics of asthma?	Asthma is characterized by reversible expiratory airflow obstruction, airway hyperreactivity, and chronic inflammation of the airways. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 361.*
Describe the treatment for status asthmaticus.	Inhaled or nebulized Beta-2 agonists are given repeatedly every 15-20 minutes, supplemental O2 is given to help maintain an oxygen saturation greater than 90%, and IV corticosteroids are administered. Due to the time it takes for corticosteroid effects to be seen, they should be initiated early in the treatment of status asthmaticus. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 186.*
What is the difference between extrinsic and intrinsic asthma?	Intrinsic asthma is caused by factors such as placement of an endotracheal tube, inhalation of irritants, exposure to cold, and exercise. Extrinsic asthma is due to triggers that activate the immune system such as inhaling allergens that stimulate the release of IgE antibodies. *Stoelting RK, Hillier SC. Pharmacology and Physiology in Anesthetic Practice. Philadelphia, PA: Lippincott Williams and Wilkins; 2006: 524.*
What are the two most common arterial blood gas findings in the asthmatic patient?	Respiratory alkalosis and hypocarbia. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 184.*

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The degree of expiratory airflow obstruction can be directly measured by what two pulmonary function tests?	The maximum mid-expiratory flow rate (FEF 25%-75%), and the forced expiratory volume in 1 second (FEV1). Patients necessitating hospitalization and treatment of asthma typically present with maximum mid-expiratory flow rates of 20% or less than normal and a FEV1 of less than 35% of normal. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 183.*
What is the primary goal of the preoperative evaluation in regards to the asthmatic patient?	To design an anesthetic that attenuates or prevents expiratory airflow obstruction. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 186.*
What undesirable effects can result from administering anticholinergic medications to an asthmatic patient?	Airway secretions can become more viscous, thus making their removal more difficult *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 187.*
What are some treatments for intraoperative bronchospasms due to asthma?	Beta-2 agonist therapy, increasing the depth of anesthetic with a volatile agent, corticosteroid administration. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 188.*
Why would a regional anesthetic be more desirable than a general anesthetic in an asthmatic patient undergoing surgery?	Manipulation of a hyperreactive airway is avoided, thus avoiding bronchoconstriction. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 187.*
Why would a deep extubation be advantageous in the asthmatic patient?	Provided there are no contraindications to deep extubation, removal of the endotracheal tube with deep anesthesia decreases the risk for bronchospasm because airway reflexes are suppressed. *Hines RL, Marschall KE. Anesthesia & Co-Existing Diseases. 6th ed. Philadelphia, PA: Elsevier-Saunders; 2012: 188.*
What factors are associated with an increased risk for perioperative complications due to asthma?	Excessive production of sputum, a prior history of asthmatic complications during the perioperative peroid, continuous or frequent corticosteriod use, recent ER visit or hospitalization due to asthma, coexisting cardiovascular disease. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 362.*

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